Joule

CellPress

COMMENTARY

Measuring Aging Stability of Perovskite Solar Cells

Michael Saliba,[1,*] Martin Stolterfoht,[3] Christian M. Wolff,[3] Dieter Neher,[3] and Antonio Abate[2,*]

Michael Saliba is a Group Leader at the Adolphe Merkle Institute in Fribourg, Switzerland. His group studies novel materials focusing on perovskites for a sustainable energy future. Michael was a Marie Curie Fellow at EPFL. He obtained his PhD at Oxford University, an MSc with the Max Planck Institute for Solid State Research, and BSc degrees in mathematics and physics from Stuttgart University. Michael was awarded the *Young Scientist Award* of the German University Association, was named one of the World's *35 Innovators Under 35* by the MIT Technology Review, and is a *Member of the Global Young Academy*.

Martin Stolterfoht is a postdoctoral research fellow in the Soft Matter Physics group at the University of Potsdam, Germany. He completed his Master degree in Physics at the University of Graz and obtained his PhD at the University of Queensland Australia in 2016 before moving to Potsdam. His research is focused on improving device efficiency through a fundamental understanding of charge transport and recombination processes in perovskite and organic solar cells.

Christian M. Wolff obtained an MSc in Physics from Ludwig-Maximilians- Universität München with a focus on spectroscopy on semiconductor nanomaterials for solar-to-fuel conversion with Prof. J. Feldmann. He joined Prof. Neher's group in 2015 investigating halide perovskites and loss mechanisms in single and multijunction solar cells.

Dieter Neher is a full professor of Soft Matter Physics at the University of Potsdam, Germany. He received his Diploma in Physics and his PhD in Chemical Physics at the Johannes Gutenberg University in Mainz, Germany. He worked as a postdoctoral research fellow at the Optical Science Center in Tuscon and the CREOL in Orlando, US. His current research focuses on the understanding and improvement of the electrical and optoelectronic properties of organic conjugated materials, hybrid organic/inorganic systems and organometallic perovskite semiconductors, and the implementation of such materials into highly efficient devices.

Antonio Abate is a group leader at the Helmholtz-Zentrum Berlin in Germany and Visiting Professor at Fuzhou University in China. His group is currently researching novel active materials and interfaces to make stable perovskite solar cells. Before his move to the Helmholtz, Antonio was leading the solar cell research at the Adolphe Merkle Institute in Switzerland. After his PhD in Politecnico di Milano in Italy, he was a Marie Skłodowska-Curie Fellow at École Polytechnique Fédérale de Lausanne and postdoctoral researcher at the University of Oxford.

Introduction

Perovskite semiconductors are among the most promising emerging semiconductor materials. They have multiple outstanding optoelectronic and material properties such as defect tolerance, a sharp band edge, and a tunable band gap in the entire visible and near-infrared range. Remarkably, perovskites can be processed by almost all standard techniques from solution processing to thermal evaporation, providing easy access for diverse scientific communities from material science, physics, chemistry, process engineering, semiconductor devices, etc. This gave rise to the notion of perovskites being a "wonder material," currently at the center stage of semiconductor research.

For photovoltaics (PVs), in an unprecedented short time, perovskite solar cells (PSCs) have surged from 3.8% efficiency in 2009 to almost 23% in 2018, on par with established technologies. PSCs could finally break the trend that high-efficiency solar cells must come at high costs, providing an inexpensive, high-efficiency, and sustainable alternative to fossil fuels.[1]

Now that PSCs are getting closer to their maximum theoretical performances, stability is the next big challenge. However, as with every new technology, perovskites have peculiar behavior for a semiconductor because of inherent ion migration, necessitating new standards for measuring stability. Developing such standards is therefore of the utmost importance to enable comparability while identifying useful figures of merit to ensure long-term progress.

In this work, we highlight the dramatic impact that hysteresis has on measuring stabilized efficiencies in PSCs. Similarly, this extends toward aging tests, where using maximum power point tracking is of utmost importance to extract reliable data. We consider the recently observed phenomenon of a reversible performance loss, which can occur in some of the highest performing PSCs to date, and how this affects the effective lifetime energy yield (LEY) of the device during day/night cycling. From this, we propose a revised T_{80} (T_{s80}) lifetime parameter, which we use to compare some of the most stable devices reported in the literature as well as newly measured

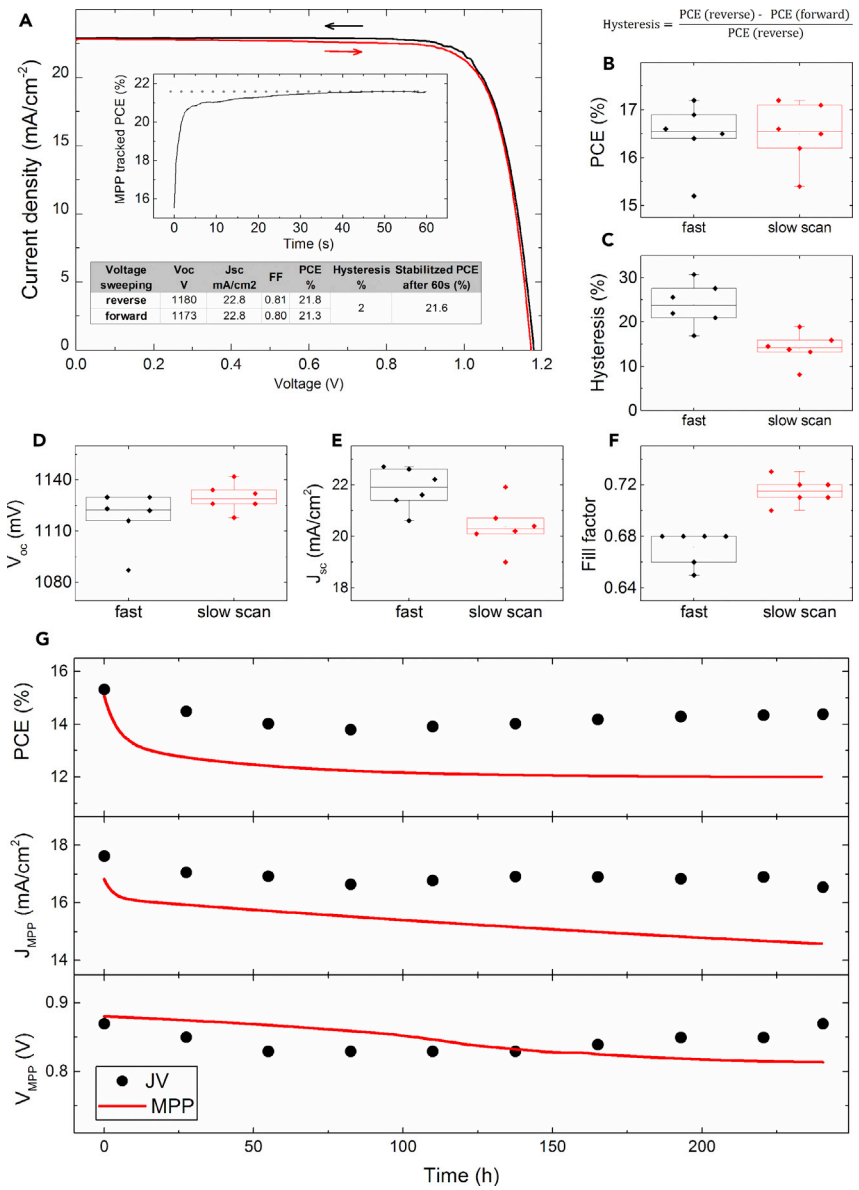


Figure 1. Hysteresis and Stabilized Power Conversion Efficiency

(A–F) JV scan (forward and reverse) with 2 rel % hysteresis (A). Inset: stabilized PCE of 21.6%.[5] Major differences arise from using fast (100 mV s^{-1}) and slow scan (10 mV s^{-1}) rates as displayed for six independent devices for (B) PCE, (C) hysteresis, (D) V_{oc}, (E) J_{sc}, and (F) FF (extracted from the reverse scan).[6] It is due to these differences that the slow scan speeds (10 mV s^{-1}) and maximum power point (MPP) tracking, as shown in the inset in (A), have become required measurements for accurate PSC performance.

(G) Tracking the maximum power output versus JV scan. PCE, current density (J_{MPP}), and voltage (V_{MPP}) at the MPP extracted from continuous MPP tracking (red lines) versus periodic JV scans collected from forward to reverse bias (black circles) for the same device.

aging data on inverted high-performance devices. With this, we provide a new figure of merit (FOM) that is more perovskite specific but also comparable with other PV materials.

We do not claim that our proposal will be the final measurement protocol for aging PSCs. However, we hope to set a starting point for an informed discussion[2–4] where aging data are collected such that long-term protocols can be developed that will support the rapid progress of PSCs toward commercialization.

Data Analysis and Discussion

Figure 1A displays a JV curve with a forward and reverse scan measured at 10 mV s^{-1} for one of the highest performing PSCs so far reported.[5] The device performance parameters, from the reverse and forward scan, are reported in the inset table. In addition to the traditional parameters such as open circuit voltage (V_{oc}), short circuit current density (J_{sc}), fill factor (FF), and maximum power conversion efficiency (PCE), the hysteresis (see formula in Figure 1) and the stabilized PCE via maximum power point (MPP) tracking are reported. Measuring PSCs in this way was established because the JV curves may exhibit a scan-rate-dependent hysteresis. This can be seen in Figures 1B–1F, where the device parameters of six identically prepared devices are shown.[6] We note that while the overall PCE can be rather similar independent of the scan rate (Figure 1B), the individual parameters (V_{oc}, J_{sc}, FF) may be scan-rate dependent as shown in Figures 1D–1F. The occurrence of hysteresis in the JV curve of PSCs has been largely addressed and slow scan rates (~10 mV s^{-1}) as well as MPP tracking have become a quasi-standard.[7]

Figure 1G displays typical aging data for mesoporous, regular PSCs. The maximum PCE is extracted from the current density (J_{MPP}) and voltage (V_{MPP}) measured at MPP (red lines) and periodic JV scans (black circles) from the same device. The J_{MPP} from the periodic JV scans becomes progressively larger compared with the same parameter extracted from MPP tracking, while the V_{MPP} fluctuates without a clear trend. From this, the PCE largely follows the J_{MPP} trend; however, it may show a significantly higher efficiency when estimated

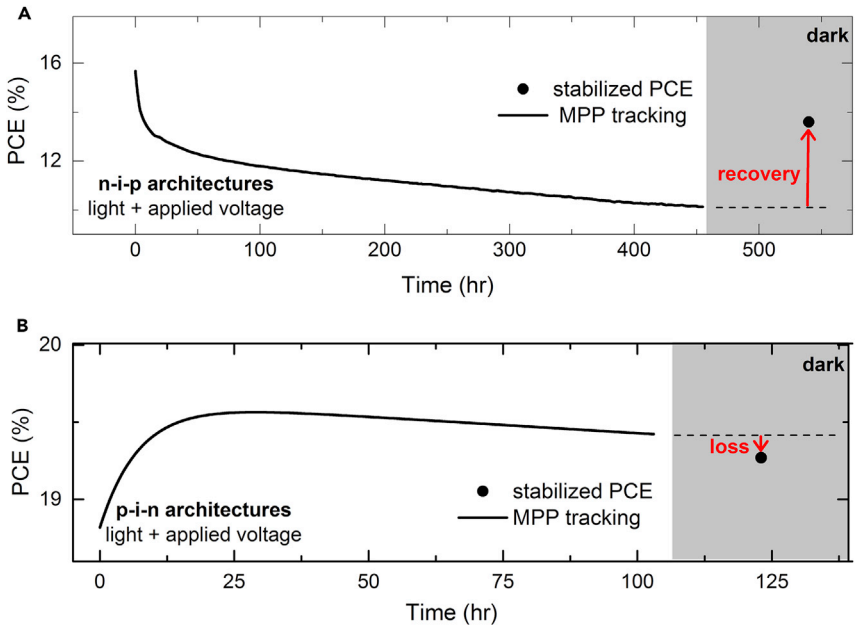


Figure 2. Initial Efficiency Losses and Dark Recovery/Losses

(A) Aging planar and mesoporous regular (or n-i-p) perovskite devices show significant efficiency recovery after resting in the dark ("reversible loss").[5,8] The device is left at open circuit in dark conditions.

(B) Planar inverted (or p-i-n) perovskite devices show an initial gain in efficiency and a loss after resting in the dark ("reversible bonus"). The analysis of aging data for inverted devices is novel and a direct consequence of applying the insights gained for regular device architectures.

from the JV curves and not the MPP tracking.[8] Such an inflated efficiency is a measurement artifact resulting from hysteresis that is becoming more severe with aging. The hysteresis exacerbates as the material degrades during device aging, most likely due to the increasing concentration of ionic defects within the perovskite.[9] Therefore, in contrast to established solar cell technologies, MPP tracking is not just a sophisticated auxiliary measurement but crucial to realistically assess long-term stability of PSCs.

One very important observation in aging high-performance PSCs is the possible presence of "reversible losses," i.e., the efficiency gain obtained by re-measuring after recovery in the dark as shown in Figure 2A. Reversible losses have been reported independently in both planar and mesoporous regular, shortened to meso regular (or n-i-p) PSCs.[5,8] Figure 2B displays the MPP trace and the stabilized PCE collected after dark recovery for a typical inverted PSC (or p-i-n) using a state-of-the-art device.[10] Interestingly, the aging behavior is symmetric to that of regular PSCs. Under MPP tracking, regular PSCs lose part of their initial efficiency, while inverted PSCs gain it. In the same way, after resting in the dark for several hours at open circuit, the regular device recovers partially, while inverted PSCs lose efficiency ("reversible bonus"). While the origin of this effect requires further extensive research, this example underlines the importance of establishing a new aging standard that systematically considers reversible energy losses/gains under relevant solar cell operating conditions, including effects of day and night cycling.

The reversible loss phenomenon has been associated with ion and defect migration within the perovskite layer as well as accumulation at the perovskite/selective contact interfaces.[11] On a different timescale, this is similar to the hysteresis phenomenon that necessitated stabilized PCE measurements (see Figure 1). Analogous to the previous discussion about stabilized PCE, the reversible ionic movement differentiates PSCs from other PV technologies, where such a reversible component is not present. Thus, the problem of measuring the efficiency of PSCs expresses itself yet again for measuring accurate stability data. This follows the trend that every PV technology has its own set of measurement requirements that are tailored to the material and the device in question.[12] In the case of PSCs, the long-term ionic movement is a unique feature that changes the accurate extraction of output parameters including stability.

We note a parallel with organic PV that exhibits a so-called "burn-in" time, i.e., non-recoverable permanent degradation occurring at the beginning of the device operation.[12] This led to revised aging protocols that excluded the initial burn-in of organic PV. Although the reversible loss component in PSCs shows some similarity to the burn-in effect in organic PVs, the underlying reasons are different (as evident from the reversibility). This sets perovskites clearly apart from previous materials, thus necessitating new measurement protocols.

Aging protocols need to have FOMs that are accessible, providing information on the long-term stability of materials and devices. Ideally, such FOMs are comparable among different technologies. One such parameter is the T_{80} test, which provides the time at which the device has degraded to 80% of the initial efficiency.[13] After this time, the device is considered to need replacement because more serious degradation mechanisms become relevant. The T_{80} test for an inorganic PV is useful scientifically and economically

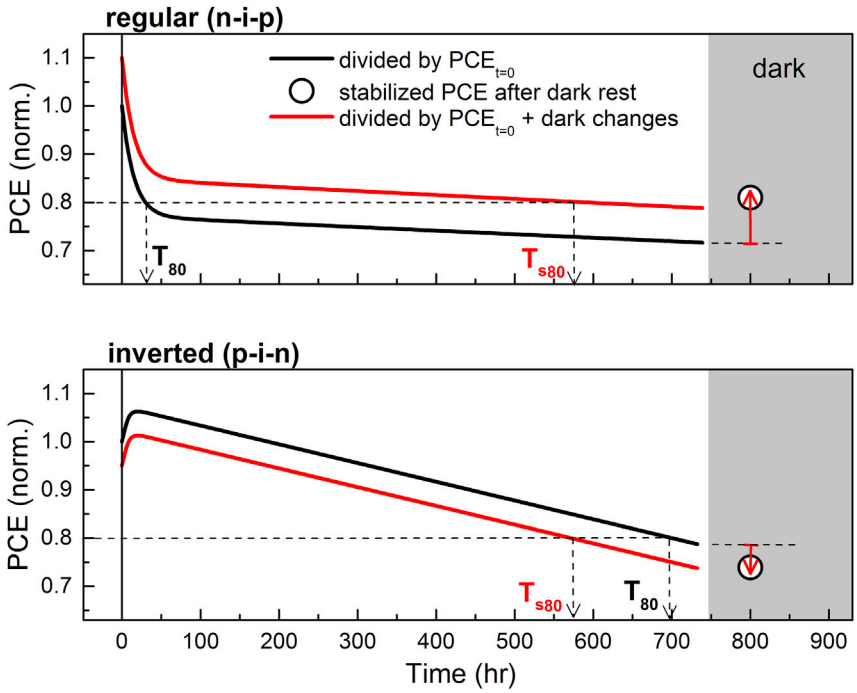


Figure 3. Aging Protocol for PSCs

Schematic of regular (top) and inverted (bottom) devices measured under MPP, including changes after dark resting. In dark conditions, the device is left at open circuit. T_{s80} (corrected T_{80}) can be extracted from the MPP traces as depicted here and used to compare the stability of perovskite with existing data from any solar cell technology.

because it helps to evaluate the LEY, which is defined as LEY = $\int_0^{T_{80}} \mathrm{PCE}(t)dt$.

The T_{80} value is well established for inorganic PVs but could not be applied as easily for organic PVs where the initial "burn-in" was removed and the component afterward was used to extract a stabilized T_{80} (called T_{s80}).[12] This means the energy that can be extracted during the burn-in period is not taken into account since it is not relevant economically. Accordingly, for organic PVs, integrating the PCE after the burn-in time can approximate the LEY. However, for PSCs, the burn-in may be reversible[5,8], and it has a significant impact on the LEY under real working conditions; i.e., day/night cycling.[11] Therefore, we should take the impact of the day/night cycling during aging into account to provide a realistic economic prospect for PSCs. Thus, we propose further aging tests should include the following key points:

- MPP tracking to avoid artificially inflated values from JV scans as described in Figure 1.
- Testing time for at least 150 hr to pass the initial transient behavior and to estimate longer-term degradation as previously reported.[6]
- Measuring the stabilized PCE after dark, resting for several hours to distinguish between reversible and irreversible processes.

Data following these guidelines permits extraction of T_{s80} for regular and inverted PSCs as schematically depicted in Figure 3. The T_{80} is defined as the time it takes for the MPP trace to decay to 80% of the initial value ($PCE_{t=0}$). After MPP tracking, the device is rested in the dark for more than 1 day. The changes in the dark are then used to renormalize the decay curve adding the reversible "loss/bonus" efficiency collected during the dark rest. Thus, the T_{s80} is defined as the T_{80} of the renormalized MPP trace as depicted in Figure 3. Accordingly, the LEY for PSCs needs to take multiple cycles into account, thus appreciating multiple reversible losses, which have a significant impact on the long-term energy yield (in contrast to previous PV materials).

The dynamics described were observed for the predominant device architectures reported in the literature. However, more subtle degradation behavior, depending on the various degradation stages, may be observed. The suggested protocol is deliberately kept general and adaptable in order to enable the detection of minute (and possibly unexpected) effects.

At this stage, major efforts are required to provide a more prescriptive, "final" stability protocol that will take years of dedicated research. We believe that the precise details such as measuring times (both in the light and dark) may well be subject to change in the future. Moreover, recording multiple day/night cycles may become very relevant. This needs careful studying to permit extraction of both the basic long-term/short-term dynamics of the device (requires longer times) and the reversible loss/bonus behavior (requires shorter times). It may well be that these long- and short-term tests become multiple, separate testing standards that are used now.[14,15] However, the guidelines provided should be considered for any measurement that investigates PSCs. Similar to efficiency measurements in PSCs, the presence or absence of hysteresis does not relieve us of collecting the forward and backward scan slowly as well as MPP tracking. While the details of these measurements may vary over time (how "slow" is a "slow scan"?), it is now clear that PSCs have an inherent risk of exhibiting "peculiar", ion-related effects. Therefore, it must be at the heart of any measurement protocol, both for efficiency and stability,

Table 1. Estimated T_{s80} from the MPP Traces of the Currently Most Stable PSCs Reported in the Literature with Different Device Architectures (Mesoporous Regular, Planar Regular, and Inverted)

Reference	Device Architecture	Initial PCE (%)	Aging Temperature and Illumination	Tracking Time (hr)	PCE Change in Dark (%)	Estimated T_{s80} (hr)
Saliba et al.[5] 2017	meso regular (n-i-p)	17	85°C, white LED, 100 mW cm^{-2}	500	+5	2,000
Tan et al.[8] 2017	planar regular (n-i-p)	20	room temperature, full solar spectrum, 100 mW cm^{-2}	500	+10	3,000
Stolterfoht et al.[10] 2017	planar inverted	16	room temperature, white LED, 100 mW cm^{-2}	170	NA[a]	9,000
Crystalline silicon modules in real working conditions					–	>50,000

We selected reports that showed more than 150 hr aging data and an initial PCE close to the state of the art. All the data reported were collected from devices measured in inert gas conditions with temperature and illumination as specified in the table.

[a]No measurement of dark recovery/loss. As reported in Figure 3, a dark loss would lower T_{s80} compared with T_{80}.

to specifically test for these unusual effects.

While there are very few datasets including cyclic aging analysis,[11] the proposed protocol can already be used on existing literature data as shown in Table 1 to provide an approximation of T_{s80} values. Here, we compare the most common device architectures, i.e., mesoporous regular, planar regular, and inverted[5,8,10] showing that, currently, PSCs (lab testing, T_{s80} < 10,000 hr) are largely outperformed by silicon (commercial modules in real working conditions, T_{s80} > 50,000 hr). Also, comparing perovskite device architectures with each other, the inverted architecture appears most stable (T_{s80} = 9,000 hr) compared with the planar (3,000 hr) and mesoporous regular (2,000 hr). Such a comparison within different perovskite architectures and with other PV materials is only possible when using a revised aging protocol, as presented here, that "standardizes" stability measurements and therefore, by definition, makes them comparable.

These considerations are guidelines toward establishing an internationally recognized aging protocol for PSCs, which will take years to analyze and develop. These first steps highlight the unique character of PSCs that need to be carefully considered when thinking about aging perovskite devices and providing comparable figures of merit to guide long-term projections.

Conclusion

PSCs have reached astonishingly high efficiency values in a short time. In the coming years, long-term stability will be one of the key tasks for research and industrial groups. Providing novel aging data for inverted devices and analyzing existing data from the most recent works, we showed that PSCs have different aging behavior from previous technologies. One very specific aspect of PSCs is that they are prone to exhibit time-dependent behavior, i.e., hysteresis in the JV scan but also reversible losses/gains in MPP tracking when kept in the dark. This requires particular attention when extracting device performance and aging metrics. We showed that such reversible losses/gains affect the LEY and thus need to be taken into account when PSCs operate in real working conditions over multiple day/night cycles. Consequently, we propose an aging protocol with a revised T_{80} (T_{s80}) lifetime parameter, which we use to compare the most stable, high-performance devices reported in the literature. Importantly, by using the aging guidelines established here, we demonstrate for the first time that inverted PSCs show a peculiar efficiency loss ("reversible bonus") when stored in the dark, which affects their T_{s80} lifetime.

We posit that this can be a starting point to develop independent, robust, aging procedures that can be the foundation for industrial employment. Without such standards from the beginning, we may be entering another "wild west" period, analogous to the hysteresis time, in which perovskite stability data cannot be compared among each other, hampering the field for years.

ACKNOWLEDGMENTS

M. Stolterfoht, C.M.W., and D.N. thank HyPerCells for funding.

1. Saliba, M., Correa-Baena, J.-P., Graetzel, M., Hagfeldt, A., and Abate, A. (2017). Perovskite solar cells from the atomic to the film level. Angew. Chem. Int. Ed. *57*, 2554–2569.

2. Yang, Y., and You, J. (2017). Make perovskite solar cells stable. Nature *544*, 155–156.

3. Saliba, M. (2018). Perovskite solar cells must come of age. Science *359*, 388–389.

4. Domanski, K., Alharbi, E.A., Hagfeldt, A., Gratzel, M., and Tress, W. (2018). Systematic investigation of the impact of operation conditions on the degradation behaviour of perovskite solar cells. Nat. Energy *3*, 61–67.

5. Saliba, M., Matsui, T., Domanski, K., Seo, J.Y., Ummadisingu, A., Zakeeruddin, S.M., Correa-Baena, J.P., Tress, W.R., Abate, A., Hagfeldt, A., and Gratzel, M. (2016). Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance. Science *354*, 206–209.

6. Abate, A., Paek, S., Giordano, F., Correa-Baena, J.P., Saliba, M., Gao, P., Matsui, T., Ko, J., Zakeeruddin, S.M., Dahmen, K.H.,

et al. (2015). Silolothiophene-linkedtriphenylamines as stable hole transporting materials for high efficiency perovskite solar cells. Energ. Environ. Sci. *8*, 2946–2953.

7. Snaith, H.J., Abate, A., Ball, J.M., Eperon, G.E., Leijtens, T., Noel, N.K., Stranks, S.D., Wang, J.T., Wojciechowski, K., and Zhang, W. (2014). Zhang, Anomalous Hysteresis in Perovskite Solar Cells. J. Phys. Chem. Lett. *5*, 1511–1515.

8. Tan, H., Jain, A., Voznyy, O., Lan, X., Garcia de Arquer, F.P., Fan, J.Z., Quintero-Bermudez, R., Yuan, M., Zhang, B., Zhao, Y., et al. (2017). Efficient and stable solution-processed planar perovskitesolar cells via contact passivation. Science *355*, 722–726.

9. Shi, L., Young, T.L., Kim, J., Sheng, Y., Wang, L., Chen, Y., Feng, Z., Keevers, M.J., Hao, X., Verlinden, P.J., et al. (2017). Accelerated Lifetime Testing of Organic–Inorganic Perovskite Solar Cells Encapsulated by Polyisobutylene. ACS Applied Materials & Interfaces *9*, 25073–25081.

10. Stolterfoht, M., Wolff, C.M., Amir, Y., Paulke, A., Perdigon-Toro, L., Caprioglio, P., and Neher, D. (2017). Approaching the fill factor Shockley-Queisser limit in stable, dopant-free triple cation perovskite solar cells. Energ. Environ. Sci. *10*, 1530–1539.

11. Domanski, K., Roose, B., Matsui, T., Saliba, M., Turren-Cruz, S.H., Correa-Baena, J.P., Carmona, C.R., Richardson, G., Foster, J.M., De Angelis, F., et al. (2017). Migration of cations inducesreversible performance losses over day/night cycling in perovskite solar cells. Energ. Environ. Sci. *10*, 604–613.

12. Peters, C.H., Sachs-Quintana, I.T., Kastrop, J.P., Beaupre, S., Leclerc, M., and McGehee, M.D. (2011). High Efficiency Polymer Solar Cells with Long Operating Lifetimes. Adv. Energy Mater. *1*, 491–494.

13. Reese, M.O., Gevorgyan, S.A., Jorgensen, M., Bundgaard, E., Kurtz, S.R., Ginley, D.S., Olson, D.C., Lloyd, M.T., Moryillo, P., Katz, E.A., et al. (2011). Consensus stability testing protocols for organic photovoltaic materials and devices. Sol. Energ. Mat. Sol. C *95*, 1253–1267.

14. Dunbar, R.B., Duck, B.C., Moriarty, T., Anderson, K.F., Duffy, N.W., Fell, C.J., Kim, J., Ho-Baillie, A., Vak, D., Duong, T., et al. (2017). How reliable are efficiency measurements of perovskite solar cells? The first inter-comparison, between two accredited and eight non-accredited laboratories. Journal of Materials Chemistry A *5*, 22542–22558.

15. Khenkin, M.V., Anoop, K.M., Visoly-Fisher, I., Galagan, Y., DiGiacomo, F., Patil, B.R., Sherafatipour, G., Turkovic, V., Rubahn, H.-G., Madsen, M., et al. (2018). Reconsidering figures of meritfor performance and stability of perovskite photovoltaics. Energ. Environ. Sci. *11*, 739–743.

[1]Adolphe Merkle Institute, University of Fribourg, Fribourg 1700, Switzerland

[2]Helmholtz-Zentrum Berlin für Materialien und Energie, Kekuléstraße 5, Berlin 12489, Germany

[3]University of Potsdam, Institute of Physics, Karl-Liebknecht-Str. 24-25, Potsdam 14476, Germany

*Correspondence: michael.saliba@unifr.ch (M.S.), antonio.abate@helmholtz-berlin.de (A.A.)

https://doi.org/10.1016/j.joule.2018.05.005

COMMENTARY

Impact of Surface Area in Evaluation of Catalyst Activity

Shengnan Sun,[1] Haiyan Li,[1] and Zhichuan J. Xu[1,2,3,4,*]

Z.J. Xu is an associate professor in the School of Materials Science and Engineering, Nanyang Technological University. He is a member of the International Society of Electrochemistry (ISE) and The Electrochemistry Society (ECS), and a Fellow of the Royal Society of Chemistry (FRSC). He served as the Vice President of ECS Singapore Section.

S. Sun is a postdoctoral research fellow in the School of Materials Science and Engineering, Nanyang Technological University. He received his PhD from the same institution.

H. Li is a PhD student in the School of Materials Science and Engineering, Nanyang Technological University. She obtained her BS degree from the Huazhong University of Science and Technology.

Electrocatalysis has been one of the hottest research fields in recent years due to the global effort in exploring sustainable energy technologies. The central effort is to develop highly active electrocatalysts for those reactions playing key roles in fuel cells, electrolyzers, and rechargeable metal-air batteries, such as oxygen evolution, oxygen reduction, and hydrogen evolution reactions. A highly active electrocatalyst should be able to promote the reaction rate of the corresponding reaction. In kinetics, the activation energy should be lowered, and, electrochemically, a lowered overpotential should be observed at a certain current density. Tafel plots are often used as a standard approach to present and compare the activities of catalysts, in which the applied potential or the overpotential at identical current density can be read. A low overpotential is usually the indicator of a good catalyst. For example, Figure S1 shows Tafel plots of oxygen evolution reaction (OER) on spinel $MnCo_2O_4$ oxides synthesized by several methods.[1] The dashed line refers to the benchmark catalyst IrO_2.[2] In general, the plots (OER catalysts) located at the upper left region are less active than those located at the lower right region. Therefore, this figure basically shows that the activity of these spinel $MnCo_2O_4$ oxides is lower than those of IrO_2.

Tafel plots are generated based on the cyclic voltammetry (CV) curves, which is the method to record the response of the working electrodes (catalysts) in current (the charge transfer at the interface of electrode and electrolyte) to the applied potential. For a typical OER measurement, the recorded current includes the OER-reaction-associated charges as well as the capacitance contribution. After capacitance correction (and optional iR correction),[2] the CV curve at the OER region (the reaction kinetic current versus the applied potential) then can be plotted as a Tafel plot in a selected potential window. It is interesting to find that the Tafel plots in many articles are different in current density. Some articles use the current density normalized by the catalyst surface area, while some of them use the current density normalized by geometric surface area of working electrodes (disk surface area) in their Tafel plots. This actually generated some confusion within the community. Which one should be used or which one makes more sense?